

Flow & Automation

Declarative process automation with Salesforce Flow – types, elements, and best practices.

• Flow Types

- **Screen Flow** – guided UI with user input.
- **Record-Triggered** – runs on create / update / delete.
- **Schedule-Triggered** – runs on a set schedule.
- **Autolaunched** – called by Apex, REST, or subflow.
- **Platform Event-Triggered** – on an event message.
- **Orchestration** – multi-step, multi-user work (Flow Orchestration).

• Record-Triggered Timing

TIMING	USE FOR
Before save	fast same-record field updates
After save	related records, email, async
Async path	callouts, slow work

Before-save updates are ~10× faster than after-save for the triggering record.

• Core Elements

- **Assignment** – set variable values.
- **Decision** – branch on conditions.
- **Loop** – iterate a collection.
- **Get / Create / Update / Delete Records** – DML.
- **Screen** – collect input (screen flows).
- **Action / Subflow** – call Apex, email, other flows.

• Resources

- **Variable** – store a value (input/output capable).
- **Formula** – computed, read-only.
- **Constant, Choice, Text Template**.
- A **collection** is a Variable with *Allow multiple values* enabled – not its own resource type.

• Entry Conditions

```
Object:      Opportunity
Trigger:     A record is updated
Condition:   StageName = 'Closed Won'
Optimize:    Actions and Related Records
```

- “Only when requirements are met” avoids needless re-runs.

• Fault Handling

- Add a **Fault path** to Get / DML / Action elements.
- Show `{!$Flow.FaultMessage}` on a screen, or log it.
- Unhandled faults roll back the whole transaction.

• Bulkification

- Flows process records in **batches** automatically.
- Never place **Get or DML inside a Loop** – assign into a collection, then act once after the loop.

• Best Practices

- Sequence multiple flows per object with **Trigger Order** and **Flow Trigger Explorer** – don’t force everything into one mega-flow.
- Extract reusable logic into **subflows**.
- Name and describe every element and resource.
- Gate with a custom permission to bypass on data loads.

• Scheduled Paths

- Add time-based paths to after-save flows.
- Offset from a date field or from the trigger fire time.
- Ideal for reminders, escalations, and follow-ups.

• Flow vs Apex vs Validation

NEED	USE
Same-record field checks	Validation Rule
Declarative logic & UI	Flow
Complex loops, callouts, bulk	Apex

Record-triggered flows can also block a save with the **Custom Error** element – like a validation rule, but with logic.

• Migrate to Flow

- **Workflow Rules** and **Process Builder** reached **end of support on Dec 31, 2025** – build all new automation in Flow.
- Convert existing rules and processes with the **Migrate to Flow** tool.

• Debug & Activate

- **Debug** replays a flow with real or test data.
- Inspect fault emails and debug logs when it errors.
- Only one **active version**; keep others as drafts.